CURRICULUM OVERVIEW

Little Coders Path
Five Courses. Real Projects.

A gentle, hands-on journey for young learners: from colourful Scratch blocks to games, AI-powered projects and block-based Python — building genuine, shareable projects the whole way, at a pace made for ages 6–11.

5	40	46	6–11
Independent courses	Lessons total	Classes with assessments	Age range

LEARNING PROGRESSION

1. **Scratch Adventures**
   Coding fundamentals, animations, stories and sprites
   MIT Scratch
2. **Scratch Game Studio**
   Advanced Scratch, game mechanics and interactive projects
   MIT Scratch
3. **AI for Kids: Smart Projects**
   Image and sound recognition with AI-powered Scratch projects
   Teachable Machine + Stretch3
4. **Python Explorers**
   Introduction to Python using visual blocks
   EduBlocks
5. **Python Creative Lab**
   Intermediate block-based Python and independent projects
   EduBlocks

Learn by playing, not by lecture. Every lesson turns a new idea into a game, story or project your child actually wants to finish.

Enrol one course at a time. Follow the recommended order or pick the course that fits your child — it is a progression, not a compulsory package.

About six months, two classes a week. 40 lessons, 5 course assessments and a final assessment make 46 live classes in total.

COURSE 1 · LESSONS 1–8 · CLASSES 1–9

Scratch Adventures

Coding fundamentals, animations, stories and sprites

MIT Scratch

OVERVIEW

Learn how code works by launching rockets, dancing aliens and telling your first animated story in MIT Scratch.

YOU WILL LEARN

- **Stage & sprites**
- **Green flag & sequencing**
- **Motion & x/y positions**
- **Looks, costumes & speech bubbles**
- **Events & keyboard interaction**
- **Sounds & loops**

BY THE END, STUDENTS CAN

- ✓ Build scripts that run in the right order
- ✓ Move, animate and talk with sprites
- ✓ Make projects that react to the keyboard
- ✓ Finish an animated story from start to end

LESSON BREAKDOWN

1. **Welcome to Scratch!**
   Project: Rocket Launch
2. **Let's Make Things Move!**
   Project: Dancing Alien
3. **Give Your Characters a Personality**
   Project: Talking Space Pet
4. **Making Things Interactive**
   Project: Control Your Rocket
5. **Sounds, Music & Effects**
   Project: Space DJ
6. **The Magic of Repetition**
   Project: Dancing Starfield
7. **Bringing Characters to Life**
   Project: Alien Animation Studio
8. **My First Animated Story**
   Project: Lost Space Puppy

★ **Course Assessment**
40 minutes · 40 marks · quick quiz plus a build-it practical project

COURSE 2 · LESSONS 1–8 · CLASSES 10–18

Scratch Game Studio

Advanced Scratch, game mechanics and interactive projects

MIT Scratch

OVERVIEW

Turn Scratch skills into real games: controls, collisions, scores, lives, levels and clones.

YOU WILL LEARN

- **Controls & boundaries**
- **Collision detection**
- **Variables, score & lives**
- **if / else & randomness**
- **Timers & levels**
- **Broadcasts & clones**

BY THE END, STUDENTS CAN

- ✓ Control a sprite with the keyboard inside a maze
- ✓ Detect collisions and keep score
- ✓ Add lives, timers and challenge
- ✓ Build a multi-level arcade game with clones

LESSON BREAKDOWN

1. **Game Controls & Movement**
   Project: Asteroid Maze
2. **Making Sprites Interact**
   Project: Catch the Falling Stars
3. **Keeping Score!**
   Project: Star Collector
4. **Making Decisions with Code**
   Project: Keycard Escape
5. **Randomness & Game Challenges**
   Project: Mystery Prize Machine
6. **Timers, Lives & Survival**
   Project: Meteor Dodge
7. **Multiple Levels & Game Scenes**
   Project: Space Mission: Three Levels
8. **Clones & Arcade Games**
   Project: Meteor Blaster

★ **Course Assessment**
40 minutes · 40 marks · quick quiz plus a build-it practical project

COURSE 3 · LESSONS 1–8 · CLASSES 19–27

AI for Kids: Smart Projects

Image and sound recognition with AI-powered Scratch projects

Generation AI Teachable Machine | Stretch3 | TM2Scratch extension

OVERVIEW

Train your own AI with a webcam and a microphone, then bring it to life in Stretch3 with the TM2Scratch extension.

YOU WILL LEARN

- **Machine learning & classes**
- **Training examples & predictions**
- **Image classification**
- **Improving a model**
- **Sound (spoken-word) classification**
- **Image & sound label events**

BY THE END, STUDENTS CAN

- ✓ Train an image model with everyday objects
- ✓ Train a model that recognises spoken words
- ✓ Connect a model to Stretch3 with TM2Scratch
- ✓ Explain how an AI learns from examples

LESSON BREAKDOWN

1. **Giga's AI Snack Buddy**
   Project: Alien Pet Café
2. **Pico's AI Space Rescue Scanner**
   Project: Pico's Rescue Mission
3. **AI Recycling Rangers**
   Project: Recycling Station
4. **AI Superhero Power-Up Lab**
   Project: Superhero Power-Up Lab
5. **AI Space Explorer Packing Mission**
   Project: Space Explorer Packing Mission
6. **AI Voice-Controlled Rocket Mission**
   Project: Voice-Controlled Rocket
7. **The Wizard's Magic Spellbook**
   Project: The Wizard's Magic Spellbook
8. **The AI Quiz Show Challenge**
   Project: The AI Quiz Show

★ **Course Assessment**
40 minutes · 40 marks · quick quiz plus a build-it practical project

COURSE 4 · LESSONS 1–8 · CLASSES 28–36

Python Explorers

Introduction to Python using visual blocks

EduBlocks

OVERVIEW

Take your first steps in Python with EduBlocks: print, draw with Turtle, repeat, calculate, ask questions and decide.

YOU WILL LEARN

- **print & output**
- **Turtle drawing**
- **Loops**
- **Variables & arithmetic**
- **input & strings**
- **if / else & random**

BY THE END, STUDENTS CAN

- ✓ Read and build Python programs with blocks
- ✓ Draw shapes and patterns with loops
- ✓ Use variables, input and decisions
- ✓ Finish a first Python quiz game

LESSON BREAKDOWN

1. **Welcome to Python!**
   Project: Mission Control Console
2. **Drawing with Python**
   Project: Rocket Badge
3. **Repeat Magic in Python**
   Project: Rainbow Galaxy
4. **Variables & Calculations**
   Project: Moon Base Supply Calculator
5. **Let the User Join In!**
   Project: Space Passport Generator
6. **Making Decisions in Python**
   Project: Oxygen Safety Checker
7. **Randomness & Smart Decisions**
   Project: Alien Treasure Machine
8. **My First Python Game**
   Project: Space Rescue Quiz

★ **Course Assessment**
40 minutes · 40 marks · quick quiz plus a build-it practical project

COURSE 5 · LESSONS 1–8 · CLASSES 37–45

Python Creative Lab

Intermediate block-based Python and independent projects

EduBlocks

OVERVIEW

Level up with lists, functions, while loops, nested loops and debugging — then build your own Python space adventure.

YOU WILL LEARN

- **Lists & indexing**
- **Looping through collections**
- **Functions & parameters**
- **while loops**
- **Nested loops**
- **Debugging & testing**

BY THE END, STUDENTS CAN

- ✓ Store and use collections of values
- ✓ Write your own functions with parameters
- ✓ Repeat until something works
- ✓ Find and fix bugs, then build a capstone

LESSON BREAKDOWN

1. **Working with Lists**
   Project: Planet Picker
2. **Loops & Collections**
   Project: Solar System Parade
3. **Creating Your Own Functions**
   Project: Star Stamp Studio
4. **Functions with Superpowers**
   Project: Custom Planet Maker
5. **Keep Going Until It Works!**
   Project: Space Password Challenge
6. **Loops Inside Loops**
   Project: Galaxy Pattern Generator
7. **Finding & Fixing Bugs**
   Project: Repair Mission Control
8. **My Python Capstone**
   Project: Space Explorer Adventure

★ **Course Assessment**
40 minutes · 40 marks · quick quiz plus a build-it practical project

The Complete Journey

Across 5 courses and 40 lessons, students progress from their first visual coding blocks to games, AI-powered projects and real Python programs — finishing every course with a project they can share.

1

Scratch Adventures

- Stage & sprites
- Green flag & sequencing
- Motion & x/y positions
- Looks, costumes & speech bubbles

2

Scratch Game Studio

- Controls & boundaries
- Collision detection
- Variables, score & lives
- if / else & randomness

3

AI for Kids: Smart Projects

- Machine learning & classes
- Training examples & predictions
- Image classification
- Improving a model

4

Python Explorers

- print & output
- Turtle drawing
- Loops
- Variables & arithmetic

5

Python Creative Lab

- Lists & indexing
- Looping through collections
- Functions & parameters
- while loops

HOW PROGRESS IS CHECKED

01 · Think in Blocks
10 multiple-choice questions · 10 marks

02 · Quick Check
5 true / false questions · 5 marks

03 · Build It!
One practical project · 25 marks, marked by the teacher

Every course ends with a 40-minute, 40-mark assessment — no negative marking. A cumulative Final Assessment closes the path, with a Scratch / AI or a Python practical project.

WHAT STUDENTS FINISH EACH COURSE WITH

Scratch Adventures	Scratch Game Studio	AI for Kids: Smart Projects	Python Explorers	Python Creative Lab
Lost Space Puppy	Meteor Blaster	The AI Quiz Show	Space Rescue Quiz	Space Explorer Adventure

TOOLS STUDENTS USE

MIT Scratch
Colourful blocks for animations, stories and games.

Stretch3
A Scratch-compatible editor with the TM2Scratch extension for AI projects.

Generation AI Teachable Machine
Train an AI model with your webcam or microphone.

EduBlocks
Drag-and-drop blocks that write real Python.

Ready to start building?
All lessons and projects are taught live by an expert Lumexa instructor.